



Department of Computer Science and Business Systems
Academic Year 2022 – 2023(Even Semester)

Degree, Semester & Branch: B.Tech., II & CSBS

Course Code & Title: AD3251 Data structures design

Name of the Faculty member: Ms.K.Usharani, AP/CSBS

Innovative Practice Description

- **Unit / Topic:** Unit 2 / Applications of Lists, Stack and Queue ADT
- **Course Outcome:** CO 2 – Implement Linear data structures for the different real world applications
- **Topic Learning Outcome:** TLO 7 - Discuss the applications of various linear data structures.
- **Activity Chosen:** Minute paper
- **Justification:**
 1. Minute paper activity will enable the students to recall the topics on their own.
 2. Better tool for brainstorming the ideas and concepts discussed in lecture.
 3. At the end of discussion on linear data structure, it is a better way to recall applications of those data structures.
- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**
 1. The students were asked to take a paper and pen.
 2. They were asked to write the applications of Lists, Stack and Queue ADT individually.
 3. The students were asked to submit the paper.
 4. This activity shows whether the students can able to understand the specific topic and their involvement the particular class.
- **CO – PO / PSO mapping:**

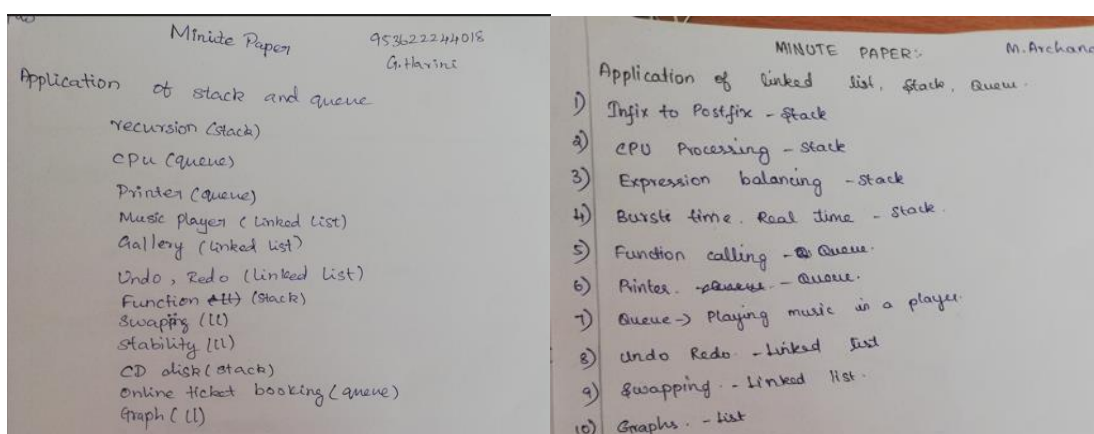
CO	PO1	PO2	PO3	PO9	PO12	PSO2
CO5	3	1	1	1	2	1

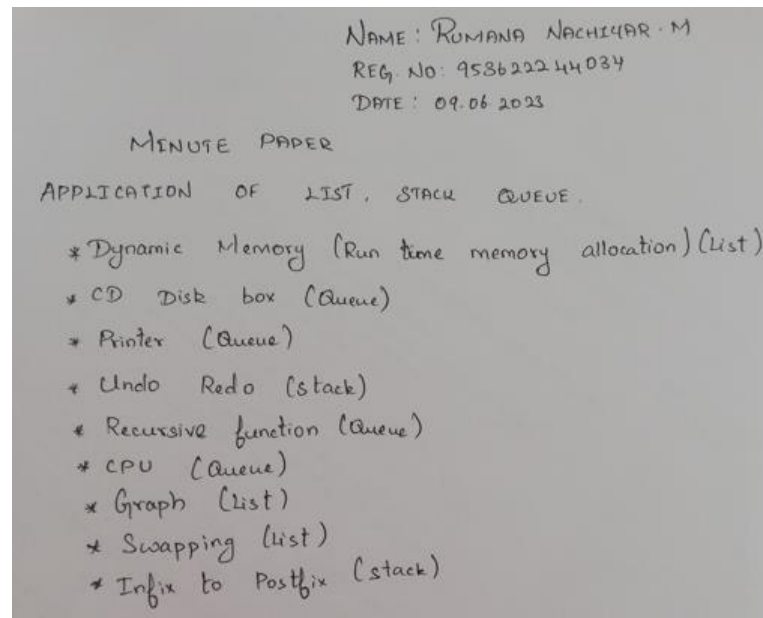
(1 – Low 2 – Moderate 3 – High)

- PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO9	PO12	PSO2
	3	1	1	1	2	1
Justification for correlation	Apply the knowledge of engineering fundamentals	Identify and analyze the applications of list, stack and queue concepts in complex engineering problems.	Design solutions for complex engineering problems by understanding the applications of linear data structures.	Function effectively as an individual by reflecting the ideas on the given topic.	Recognize the need and have the preparation and ability to engage in independent learning.	Motivates Continuous learning to provide the solution to real world business problems.

- Images / Screenshot of the practice:





- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

Students felt that they can able to recall the applications of linear data structures. And this activity helped them to analyze their level of focus on the classroom.

- ❖ **Benefit of the practice:**

1. Students were actively participated in this activity.
2. It increases students focus in class and also improves the student's writing skill and analyzing ability on the topic.

- ❖ **Challenges faced in implementation:**

Some students felt difficult to recall the topics on their own.

References:

1. Michael T. Goodrich, Roberto Tamassia, and Michael H. Goldwasser, "Data Structures and Algorithms in Python" (An Indian Adaptation), Wiley, 2021.
2. Narasimha Karumanchi, "Data Structures and Algorithmic Thinking with Python" Careermonk, 2015.